AI-Based Pneumonia Detection System

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Implementation

A deep dive into how Python-generated AI predictions flow through a structured data pipeline into an interactive Power BI dashboard for clinical decision support.

PHASE 2 OVERVIEW

Phase 2 Methodology at a Glance

Data Generation

Python simulation creates raw datasets

Data Processing

Pandas pipeline cleans and writes CSVs

AIPrediction

Model outputs actual, predicted, confidence

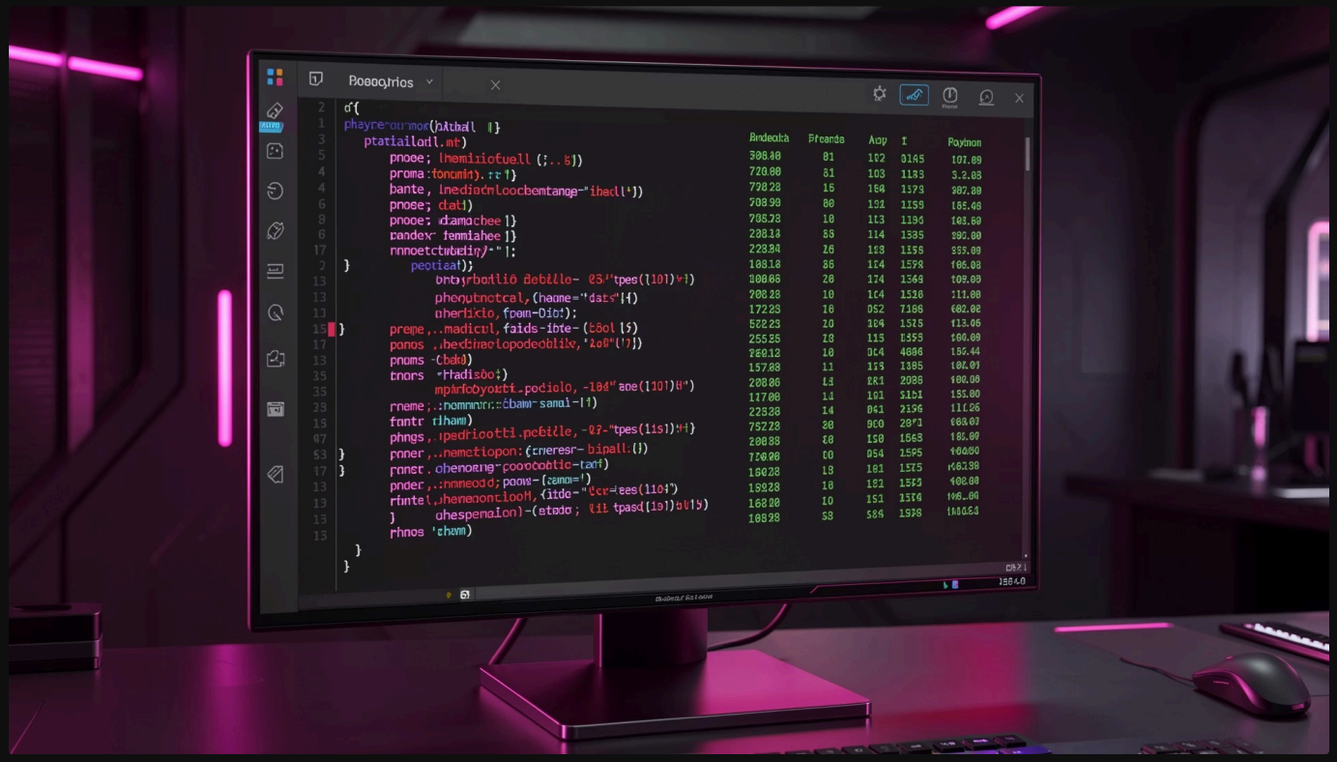
Visualization

Power BI dashboard for insights

Phase 2 shifts from theory to execution — every component is built, tested, and connected in a working end-to-end pipeline.

DATA GENERATION

Generating the Dataset with Python



Python scripts simulate realistic clinical prediction data, producing hundreds of labeled patient records. The `random` and `pandas` libraries are used to generate and export structured CSV files.

Volume

500–1,000 synthetic patient records per run

Format

Exported as `.CSV` for downstream processing

Reproducibility

Seeded random generation ensures consistent results

DATASET FIELDS

Understanding the Three Core Fields

Actual

The ground-truth label — **1 = Pneumonia**, **0 = Normal**. Represents the clinically verified diagnosis used to evaluate model accuracy.

Predicted

The AI model's output classification for each patient record. Compared against Actual to measure correctness and surface errors.

Confidence

A score from **0.0 to 1.0** indicating how certain the model is about its prediction. High confidence + correct prediction = reliable output.

Tools & Technologies

Python

Corescripting language for data generation, model simulation, and CSV export automation.

Pandas

Handlesdata manipulation — cleaning, filtering, aggregating, and structuring records before export.

Power BI

Microsoft'sBI tool renders the CSV data into interactive charts, KPI cards, and filterable dashboards.

STEP-BY-STEP PROCESS

How the System Works

01

Run Python Script

Execute the data generation script to simulate patient records with Actual labels and AI Predicted values.

02

Apply Pandas Processing

Clean and validate data — handle nulls, normalize Confidence scores, and compute accuracy metrics.

03

Export to CSV

Save the finalized dataset as a structured .csv file ready for import.

04

Import into Power BI

Load the CSV, define relationships, and build visuals — confusion matrix, confidence histograms, and accuracy KPIs.

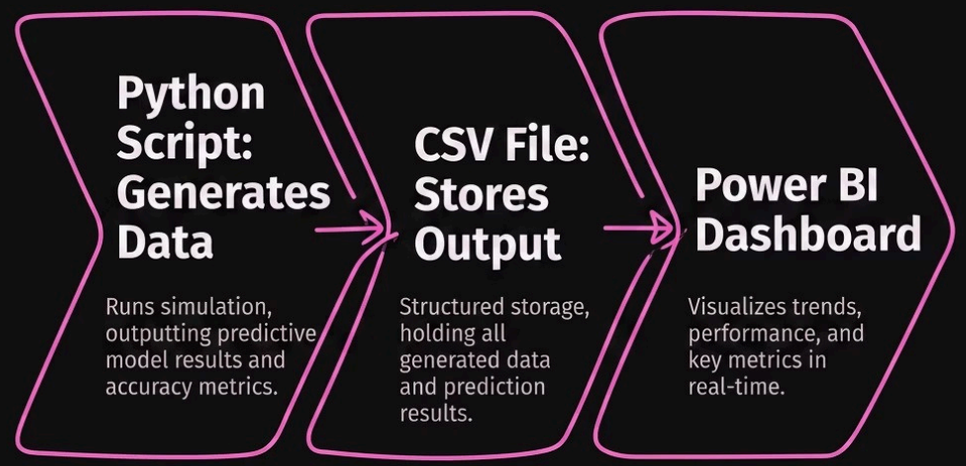
05

Publish & Interact

Share the live dashboard for filtering by date, confidence threshold, or prediction outcome.

ARCHITECTURE

System Architecture



The architecture is deliberately **lightweight and modular**. Each layer is independent — Python can be swapped for a live model, and Power BI can be replaced with any BI tool that accepts CSV input.

No database or backend server is required in Phase 2 — the CSV acts as the data store, keeping the pipeline simple and portable.

DATA PIPELINE

Data Processing& Storage

Cleaning

Pandasremovesduplicate rows, fills missing values, and enforces data types — ensuring every record is valid before export.

Aggregation

Summarycolumnsare computed: total correct predictions, average confidence per class, and per-patient accuracy flags.

Storage

Final CSV isversion-stamped with a timestamp and stored locally. Power BI reads directly from the file path for refresh.

AI ROLE

Role of AI in Prediction

In Phase 2, AI is represented through **simulated model outputs** that mirror the behavior of a trained classifier (e.g., CNN or Random Forest). The system captures how a real AI model would behave — including uncertainty, misclassification, and confidence variation.

- **Binary classification:** Pneumonia vs. Normal
- **Confidence scoring:** Probabilistic output per record
- **Error analysis:** False positives and negatives tracked

Why Simulate?

Phase 2 validates the full pipeline — data flow, dashboard logic, and visual design — before integrating a live trained model in Phase 3.

CHALLENGES

Challenges Faced in Phase 2

Realistic Data Simulation

Generating synthetic data that mirrors real-world class imbalance (more Normal than Pneumonia cases) required careful probability weighting.

Power BI Data Refresh

Configuring automatic refresh from a local CSV path required matching file directory permissions and Power BI Desktop settings.

Confidence Calibration

Ensuring simulated confidence scores were not uniformly distributed — real models show clustering near 0 and 1, not a flat spread.

Dashboard Clarity

Designing visuals that communicate both accuracy and confidence without overwhelming non-technical stakeholders took several iterations.

The End

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